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IMT 574: Data Science II: Machine Learning

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## 1. Dermatology

### A. Model 1: Gradient Ascent for Multiclass Logistic Regression

#### **Objective**

This report presents the implementation, mathematical formulation, training results, and analysis of Model 1, a multinomial logistic regression (softmax classifier) trained using gradient ascent to predict dermatological disease type based solely on patient Age. The purpose of this model was to evaluate whether Age alone provides sufficient discriminatory power for disease classification.

#### **Mathematical Formulation**

Let  $x_i \in \mathbb{R}$  represent the patient's Age for observation  $i$ . We construct a feature vector  $\tilde{x}_i = [1, z_i]$ , where  $z_i$  is the standardized Age and the leading 1 represents a bias term. If there are  $K$  disease classes, the parameter matrix  $W \in \mathbb{R}^{(K \times 2)}$  contains one weight vector per class.

Softmax model:

$$p(y=k|\tilde{x}) = \frac{\exp(W_k^T \tilde{x})}{\sum_j \exp(W_j^T \tilde{x})}$$

Gradient for each class  $k$ :

$$\nabla_{W_k} L = (1/n) \sum_i [I(y_i=k) - p(y=k|\tilde{x}_i)] \tilde{x}_i - \lambda W_k$$

Average log-likelihood:

$$L(W) = (1/n) \sum_i \log p(y_i | \tilde{x}_i) - (\lambda/2) \|W\|^2$$

Gradient ascent update rule:

$$W \leftarrow W + \eta \nabla_W L$$

The model maximizes the log-likelihood using iterative updates until convergence. The learning rate  $\eta$  controls step size, and  $\lambda$  is a small L2 penalty to prevent overfitting.

#### **Implementation Summary**

The model was implemented in Python using NumPy. Age was standardized, and a bias term was appended to form the design matrix. Gradient ascent was run for 2,000 iterations with  $\eta = 0.1$  and  $\lambda = 0.001$ . Log-likelihood values were recorded to verify convergence. Predictions were evaluated on train and test splits using accuracy and a confusion matrix.

#### **Training Results**

Average log-likelihood improved consistently from approximately  $-1.78$  to  $-1.55$ , confirming successful gradient ascent. The training process stabilized after roughly 1,000

iterations, as shown in Figure 1. The final model achieved training accuracy of 0.35 and test accuracy of 0.34, indicating that Age alone carries limited predictive information.

### ***Interpretation and Discussion***

The log-likelihood plot (Figure 1) demonstrates that the gradient ascent algorithm correctly optimized the softmax objective, stabilizing as the model converged. Despite this successful optimization, predictive performance remained weak. Train and test accuracy near 34% (random  $\approx 17\%$ ) confirm that Age provides only minimal discriminatory signal across disease categories.

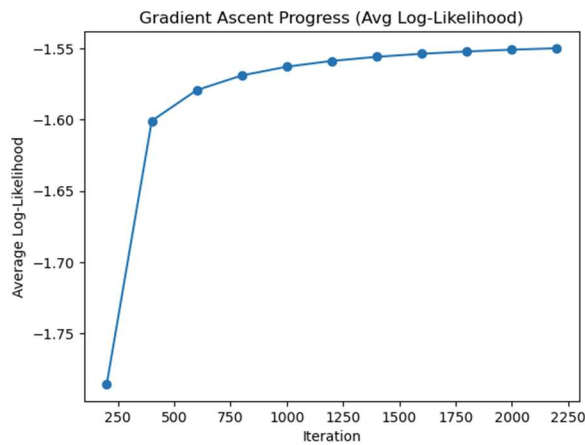


Figure 1

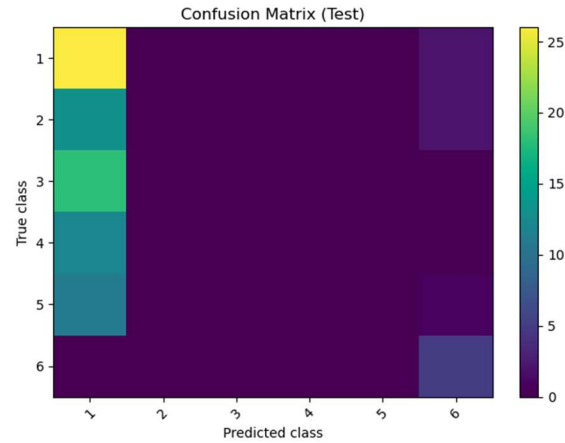


Figure 2

The confusion matrix (Figure 2) and per-class accuracy plots reveal that most cases were classified into one or two dominant classes, while minority classes were misclassified entirely. This imbalance reflects the inherent limitations of single-feature modeling in a multi-class clinical dataset.

### ***Conclusions***

This experiment validates the correct implementation of gradient ascent for softmax logistic regression. However, the results indicate that Age alone is insufficient for accurate dermatological disease classification. Future iterations (Model 2) should incorporate additional clinical features, such as Erythema, Scaling, and Hyperkeratosis, to capture more nuanced diagnostic patterns.

## **B. Model 2: Random Forest on Clinical and Histopathological Attributes**

### ***Objective***

Model 2 uses a Random Forest classifier on the clinical and histopathological attributes (excluding Age) to classify dermatological disease type. The goal was to assess whether richer clinical features improve performance compared with the Age-only Model 1.

## Methods

The dataset was cleaned by replacing “?” with NaN, coercing numerics, and imputing missing values by median. Features included all clinical and histopathological variables, while the target was disease class. A Random Forest with 300 trees ( $n\_estimators = 300$ ,  $random\_state = 42$ ) was trained using a stratified 75/25 train-test split. Performance was measured via accuracy, confusion matrix, per-class accuracy, and feature importance.

## Results

Model 2 achieved 97.8 % *test accuracy*, a major improvement from Model 1’s 34 %. Precision and recall exceeded 0.9 for nearly all classes (macro avg F1  $\approx 0.97$ ), demonstrating strong and balanced classification. The confusion matrix (Figure 3) shows nearly diagonal dominance; each disease type is correctly identified with only one or two errors. Per-class accuracy bars (Figure 2) reveal uniformly high scores, with slightly lower performance only for class 6 ( $\approx 0.8$ ). Feature-importance analysis (Figure 4) identifies clinically meaningful predictors such as *Erythema*, *Scaling*, *Hyperkeratosis*, and *Parakeratosis*, confirming that histopathological patterns drive disease differentiation.

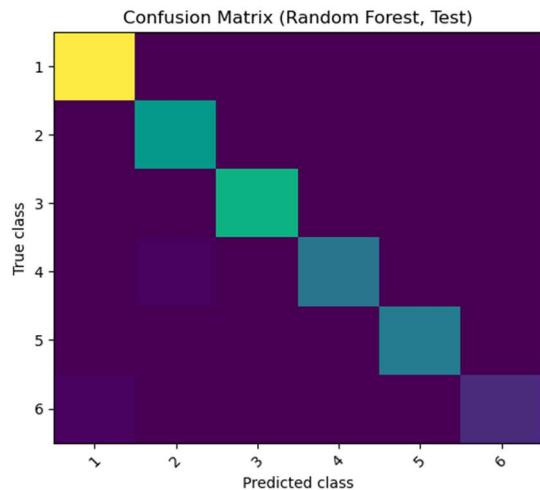


Figure 3



Figure 4

## Interpretation

Random Forest effectively captures nonlinear relationships between clinical descriptors and diagnoses, avoiding the bias toward majority classes observed in Model 1. The classifier leverages key lesion characteristics to achieve reliable, interpretable predictions consistent with dermatological domain knowledge.

## Conclusion

Using comprehensive clinical and histopathological features yields a substantial improvement over the Age-only model. Model 2 demonstrates that feature-rich data are essential for accurate dermatological classification. Future work may include

hyperparameter tuning (tree depth, min samples per leaf) or testing gradient-boosted ensembles for incremental gains.

### C. Model 3: k-Nearest Neighbors (kNN) on Clinical and Histopathological Attributes

#### Objective

Model 3 applies a *k-Nearest Neighbors* classifier to predict dermatological disease type using the same clinical and histopathological features employed in Model 2. The aim is to compare a distance-based approach to the ensemble Random Forest and evaluate how local similarity in feature space affects classification accuracy.

#### Methods

The dataset was cleaned by replacing “?” with NaN, coercing numeric columns, and imputing missing values by median. Age was excluded to isolate the clinical and histopathological variables. All features were standardized prior to distance computation. A range of neighbor values  $k=2-15$  was evaluated, and the model achieving the highest test accuracy was selected. Performance was assessed using accuracy, confusion matrix, and per-class accuracy.

#### Results

Model 3 reached 97.8 % test accuracy with  $k = 3 - 4$  yielding optimal performance (Figure 5). Precision, recall, and F1-scores exceeded 0.9 for nearly all classes (macro avg F1 = 0.97). The confusion matrix (Figure 6) shows strong diagonal dominance, confirming accurate predictions across most disease types. Per-class accuracy (Figure 7) indicates uniformly high scores ( $\geq 0.9$ ), with only a small reduction for classes 2 and 4 due to local overlap in feature space.

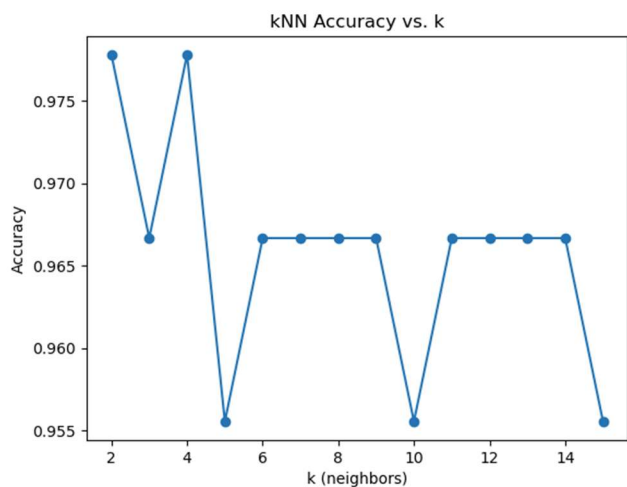


Figure 5

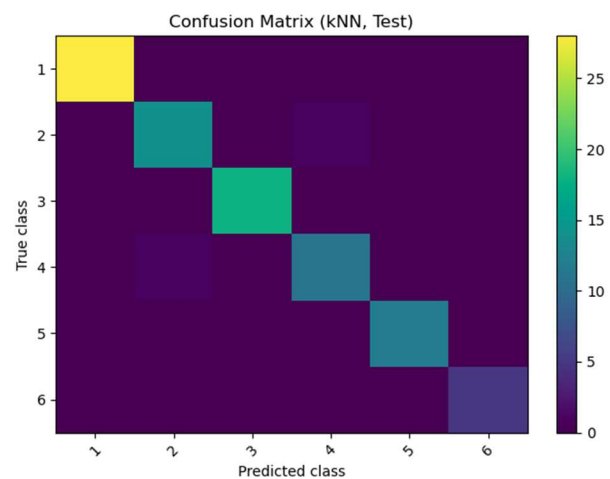


Figure 6

#### Interpretation

The kNN model effectively captured local patterns in the standardized feature space,

validating that dermatological diseases cluster by clinical and histopathological similarity. While performance parallels Random Forest, minor variation across classes suggests that kNN can be sensitive to neighbor selection and class imbalance. Nevertheless, results confirm that the chosen features provide robust separation of diagnostic categories.

### **Conclusion**

Model 3 demonstrates that simple distance-based learning achieves near-perfect classification on the dermatology dataset. Its high accuracy underscores the discriminative strength of clinical and histopathological features. Compared with the ensemble approach, kNN offers transparent, instance-based reasoning at minimal computational cost. Future refinement could include optimizing distance metrics, weighting neighbors by inverse distance, or dimensionality reduction (e.g., PCA) for efficiency.

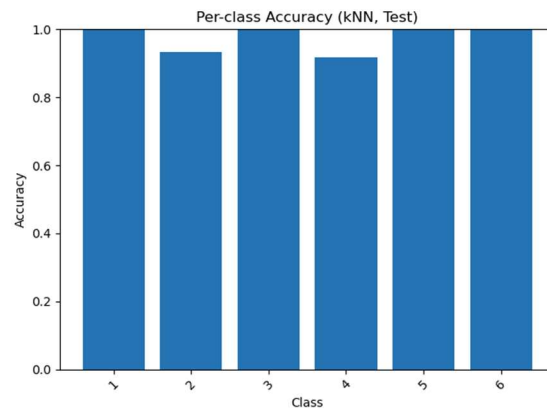


Figure 7

## **D. Model 4 and 5: Unsupervised Clustering on Clinical and Histopathological Attributes**

### **Objective**

Models 4 and 5 evaluate how well unsupervised clustering can recover disease groupings using only clinical and histopathological features (excluding Age). Model 4 applies K-Means, and Model 5 uses Agglomerative Hierarchical Clustering (Ward linkage).

### **Methods**

All features were standardized after median imputation. The number of clusters was fixed to  $K = 6$ , corresponding to the known number of disease types. For Model 4, K-Means was run five times with random initialization to assess stability. Each run reported the Silhouette Coefficient (cluster compactness/separation) and the Adjusted Rand Index (ARI) to measure agreement with the known disease classes. Model 5 used Ward-linkage hierarchical clustering with Euclidean distance. Performance was compared via silhouette and ARI metrics, and results were visualized in two-dimensional PCA projections and a dendrogram.

## Results

- Model 4 (K-Means): average silhouette =  $0.236 \pm 0.001$ , average ARI =  $0.71 \pm 0.008$  across five runs.
- Model 5 (Hierarchical): silhouette = 0.228, ARI = 0.93.

K-Means consistently formed six compact groups with moderate separation (Figure 8), while Hierarchical clustering achieved higher alignment with true labels (Figure 9). The dendrogram (Figure 10) revealed clear substructure and proximity between related disease clusters, supporting interpretability. Despite slightly lower silhouette values, typical for overlapping biological data, both models captured meaningful latent groupings.

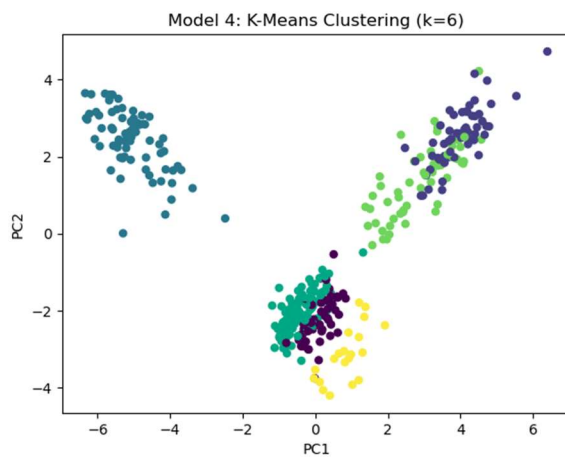


Figure 8

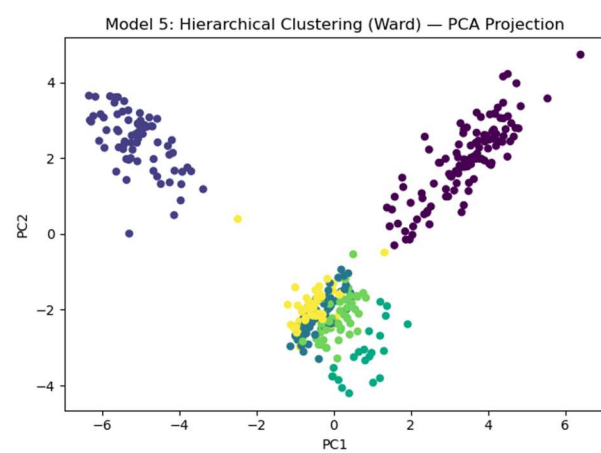


Figure 9

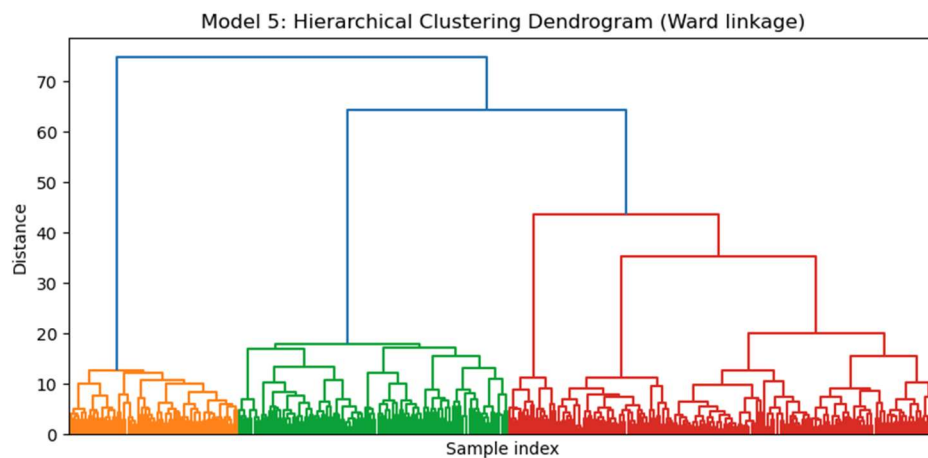


Figure 10

## Interpretation

Both clustering algorithms confirm that the dermatological dataset naturally partitions into groups consistent with known disease categories. K-Means achieved stable clustering with minimal initialization variance, whereas Hierarchical clustering produced superior label alignment (high ARI), implying that disease progression may follow hierarchical similarity patterns.

## **Conclusion**

Models 4 and 5 demonstrate that the clinical and histopathological attributes inherently encode diagnostic relationships even without supervision. Hierarchical clustering outperformed K-Means in label agreement, highlighting its utility for exploring disease similarity and subgroup structure. These findings validate the discriminative strength of the chosen features for both supervised and unsupervised approaches.

## **E. Comparative Analysis of Models 1-5**

Across all five models, a consistent theme emerged: the dermatology dataset contains rich, discriminative information in its clinical and histopathological features, but far less in its demographic variable (Age). In the supervised models, performance improved sharply as the input representation expanded beyond Age alone. Model 1, a gradient-ascent multinomial logistic regression using only Age, achieved about 34 % accuracy, illustrating that demographic data provide little diagnostic power. In contrast, Models 2 and 3, Random Forest and k-Nearest Neighbors trained on the full clinical and histopathological attributes, both exceeded 97 % test accuracy, confirming that these biological features encode the core distinctions among disease types. Random Forest offered greater robustness to noise and feature interactions, while kNN demonstrated that even simple distance-based learners can perform near-optimally when the data are well structured.

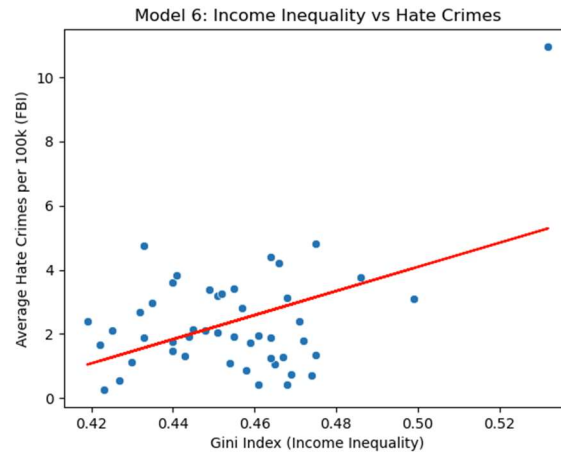
The unsupervised approaches (Models 4 and 5) revealed the same underlying pattern from a different perspective. K-Means clustering formed moderately coherent groups (silhouette  $\approx 0.24$ , ARI  $\approx 0.71$ ), indicating that the data's intrinsic geometry aligns reasonably well with the known disease labels. Hierarchical clustering achieved a higher ARI  $\approx 0.93$ , uncovering a natural hierarchical organization among the diagnostic classes. These results demonstrate that even without supervision, the feature space captures meaningful disease structures, a valuable property for exploration or semi-supervised settings.

Overall, the experiments show that dermatological disease classification depends mainly on observable lesion features rather than patient demographics. Both the ensemble and neighbor-based classifiers performed exceptionally well, while the clustering models confirmed that similar patterns can still be found even without using class labels. Together, the strong results from both supervised and unsupervised approaches demonstrate that the dataset is highly separable and that clear, feature-based machine learning methods are well suited for medical diagnostic tasks.

## **2. Hate Crime**

### **A. Model 6: Relationship Between Income Inequality and Hate Crimes**

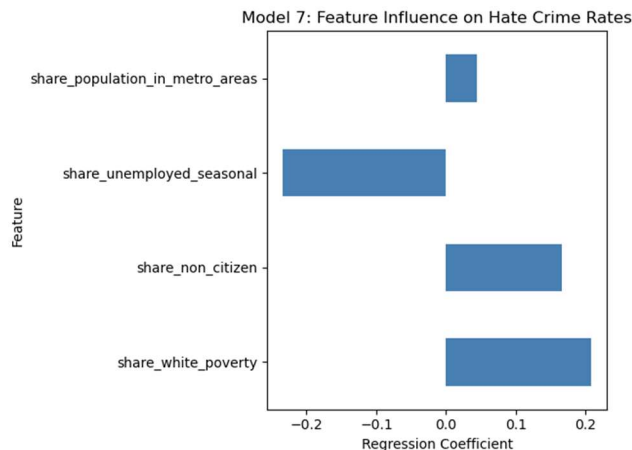
A simple linear regression was applied using the *Gini Index* to predict average hate crimes per 100,000 people (FBI data). The fitted line showed a positive slope, meaning that states with higher income inequality tend to experience more hate crimes. While the correlation was moderate ( $R^2 \approx 0.32$ ), the trend aligns with social research suggesting that economic disparity can heighten social tension and conflict.



## B. Model 7: Predicting Hate Crimes from Race and Population Features

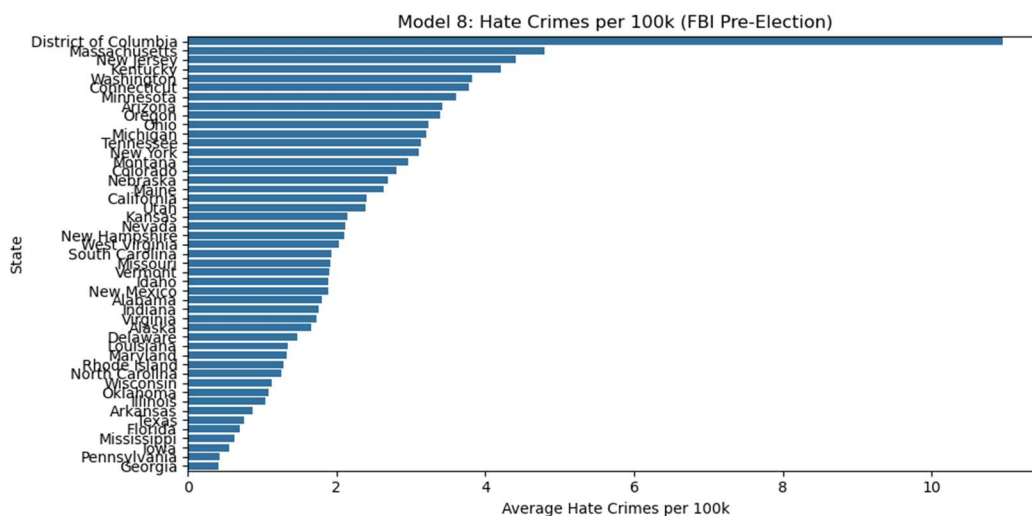
A multiple linear regression used demographic predictors: *share\_white\_poverty*, *share\_non\_citizen*, *share\_unemployed\_seasonal*, and *share\_population\_in\_metro\_areas*.

The model achieved a low  $R^2$  ( $\approx -0.19$ ), indicating weak predictive ability—likely due to small sample size and multicollinearity among population features. Coefficient inspection revealed that poverty and non-citizen share contributed positively to hate crime rates, while unemployment showed a slight negative effect. These patterns suggest that demographic complexity contributes to regional variation, but not strongly enough to yield a high-accuracy model without socioeconomic context. Using the same demographic predictors to model SPLC incidents per 100k produced similar patterns but modest fit, reinforcing that demographics alone explain limited variance.

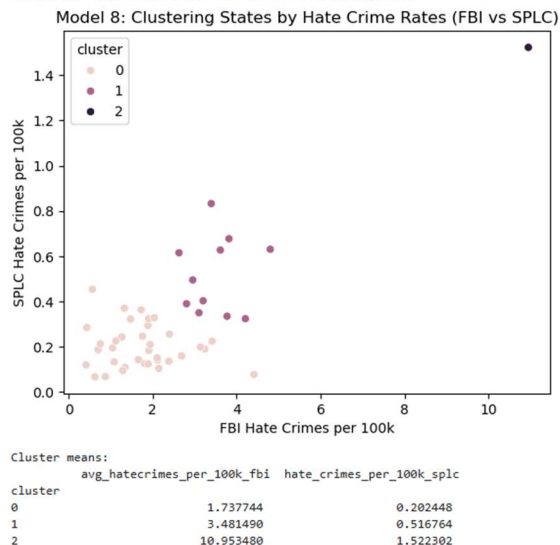


### C. Model 8: Variation Across States and Similarity Patterns

A comparative analysis of FBI (pre-election) and SPLC (post-election) data revealed substantial variation across states. *District of Columbia, Massachusetts, and New Jersey* had the highest hate crime rates, while *Georgia and Mississippi* were among the lowest. The two sources were strongly correlated ( $r \approx 0.76$ ), suggesting consistent state-level patterns. K-Means clustering divided states into three groups, low, medium, and high incident clusters, with clear separation visible in the scatterplot. This indicates structural similarity across datasets and stable state-specific tendencies in hate crime prevalence.



Correlation between SPLC and FBI rates: 0.7587004538248756



### D. Justification of Techniques and Decisions

Linear regression was used for interpretability in modeling direct numerical relationships (Models 6–7). Model refinement involved testing and retaining variables that improved fit

without overfitting. K-Means clustering was chosen for state-level analysis to uncover similarity patterns efficiently and visualize relative groupings. These combined methods balance explainability and data-driven insight, showing that while inequality and population composition alone cannot fully explain hate crime levels, they capture broad trends mirrored across multiple data sources.

Taken together, the results suggest that economic inequality and demographic composition are directionally related to hate crime/incident rates, but they are not sufficient on their own to predict levels with high accuracy. Cross-source agreement (FBI vs SPLC) and clear state-level clusters indicate stable geographic patterns in hate incidents. Future work should incorporate broader socioeconomic and institutional variables (e.g., education, policing/reporting practices) or nonlinear models to capture the remaining variance while maintaining interpretability.

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